

# Engineering Human Capital: Long-Run Effects of Wartime Technical Training \*

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## Abstract

This paper examines the enduring impact of the Engineering, Science, and Management War Training (ESMWT) program (1940-45) on the growth of engineers and scientists. To identify the effect of this training, the analysis leverages residual variation in ESMWT intensity across commuting zones, which stems from prediction errors made by local areas regarding imminent labor shortages during WWII, conditional on other WWII-related factors. The results indicate that this training led to a lasting increase in the number of engineers and STEM workers. Further analyses suggest that it also boosted interest in STEM education and led to increased post-graduate training after WWII. Localized spillovers in defense-related industries, driven by a temporary wartime demand shock, appear to have played a limited role in these changes in the occupational structure.

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# 1. Introduction

To what extent a government-funded education can increase the supply of engineers? To answer this question, this paper investigates a program known as the Engineering, Science, and Management War Training (ESMWT) program, primarily focusing on STEM training. In the wake of World War II, the United States found itself facing a critical shortage of skilled workers versed in the realms of science and engineering. To address this dire need, the U.S. government joined forces with universities and colleges, providing training to 1.8 million individuals between 1940 and 1945. ESMWT significantly distinguished itself from other educational initiatives, like the GI Bill, which primarily supported degree-granting programs in colleges. Instead, the ESMWT program offered a compact 12-16 week educational experience tailored to meet the needs of the wartime economy. In essence, it resembled an extension or night school program, aiming to swiftly impart essential knowledge essential for wartime production. It's important to note that the program's focus did not encompass the provision of comprehensive education in fields such as engineering, science, or business. Rather, its central mission was to provide specialized training that effectively catered to the distinct requirements of defense and war-related industries (Armsby, 1946, p.20).

While traditional training for professional engineers, chemists, and physicists spans 7 to 8 years, specific tasks within these fields can be managed by individuals with a more focused general education and specialized skill training. This program, unlike graduate-level scientific training, prioritized preparing individuals for roles like draftsmen, lab technicians, inspectors, and testers, enabling them to excel in specialized tasks without extensive grounding in basic science. For instance, a structural engineer could transition into a stress analyst role in aircraft design through concentrated specialized training.

Despite many trainees lacking college diplomas and prior science or engineering experience, evidence suggests this training had a significant impact. It rekindled interest in adult education, fostering a widespread recognition of effective teaching's transformative potential. The program demonstrated that adults could acquire new knowledge through textbooks and lectures, even in areas they had previously deemed challenging. It broadened horizons and enriched the lives of many beyond traditional school age (Armsby, 1946, p.65).

However, some evidence suggests that the availability of accelerated education,

like community college, can deter individuals from attending four-year colleges and encouraging them to opt for community colleges instead (Mountjoy, 2022). This raises questions about the program's impact on the local labor market in the long run and whether it genuinely increased the density of skilled workers. Furthermore, this program was only in effect during the wartime, which raises questions about the long-term persistence of its effects.

To address these questions, we systematically investigate the consequences of this training program on local labor markets and explore the underlying mechanisms. A key challenge in establishing causality arises from the potential concentration of training in areas with higher initial education levels, which may independently drive growth.

To overcome this challenge, we exploit variation in training allocation, driven by the unpredictable nature of wartime supply contracts. Regional War Manpower Commission directors, working alongside industrial executives and local universities, identified future labor shortages several months ahead and determined the training needs in local areas. The War Manpower Commission's ES-270 reports offered a survey of labor shortages in local labor markets. As noted by Bixler (1944), the share of trainees correlated with labor shortages identified in ES-270 reports and also with wartime supply contracts. The predictions of labor shortages were periodically updated throughout World War II, and the allocation of ESMWT training was responsive to these evolving forecasts.

However, industrial executives often lacked precise insights into their industries' training and personnel needs, hindering their ability to anticipate requirements beyond immediate operations. These factors included local war contracts, production types, labor markets, project sizes, or Selective Service policies that exempted certain groups of people from military conscription. Moreover, many industrial executives were unaware of the existence of ESMWT program (Armsby, 1946, p.101).

This suggests that the allocation of more training to certain local areas is not associated with their economic potential after controlling for factors associated with wartime labor scarcity, such as actual wartime supply contracts and the share of WWII veterans. Rather, it is driven by inaccurate predictions made by the program's regional directors regarding their actual training needs a few months in advance.

To validate this identifying assumption, we conduct a balance test across commuting zones, considering ESMWT training, both with and without the inclusion

of WWII-related characteristics. While the intensity of ESMWT training initially exhibited associations with economic indicators such as the proportion of college-educated workers, STEM workers, engineers, manufacturing value added, and wages, these imbalances disappeared when we introduced controls for WWII characteristics.

Building on this identifying assumption, our paper proceeds to assess the long-term repercussions of ESMWT training across various levels: commuting zones, cohorts, individuals, and universities. We draw upon data from diverse sources, including the census, patent records, public opinion polls, and graduate dissertations. Our findings reveal that this training led to a higher proportion of engineers and STEM workers, and this effect endured until at least the year 2000, with no discernible pre-existing trend before 1940.

Moreover, an analysis of a public opinion poll conducted in 1948 highlights that individuals residing in states with more intensive training expressed heightened interest in STEM subjects while exhibiting reduced interest in agriculture and home economics. This suggests a shift in people’s academic interests from agriculture and home economics towards STEM disciplines.

To ascertain whether this heightened interest in STEM fields translated into actual academic pursuit, we examine the number of master’s and doctoral dissertations in STEM disciplines across universities and colleges involved in the training program. We find that universities that trained a greater number of workers relative to their pre-war count of engineering students produced more graduate students majoring in STEM fields.

This paper effectively rules out alternative explanations. It is possible that the temporary demand surge during WWII gave rise to new industries related to war production, leading to sustained agglomeration in these sectors (Gross and Sampat, 2023; Kantor and Whalley, 2023; Schweiger et al., 2022). To explore this aspect, we conduct two additional analyses. First, we use patent data to investigate whether there was an increase in the number of patents per capita. Although a temporary uptick is evident from 1941 to 1950, this effect waned shortly thereafter. Second, we leverage census data to identify manufacturing sectors that experienced the most substantial growth between 1940 and 1950. We then examine whether ESMWT training contributed to an expanded workforce in these industries. Our findings reveal that the growth of defense-related sectors decelerated shortly after World War II,

with a significant reduction in effect for cohorts born after the 1960s. Taken together, these results suggest that the effect of ESMWT on the increase of STEM workers is driven by an increase in supply rather than by an increase in demand.

## 1.1 Related literature

This paper makes contributions to several strands of economic literature. It investigates the determinants of the growth of the engineering occupation, a factor that has been shown to play a pivotal role in driving long-term economic growth in various studies (Hanlon, 2022; Maloney and Valencia Caicedo, 2022; Murphy et al., 1991; Peri et al., 2015). To the extent that the supply of STEM workers remains relatively inelastic, demand-side policies such as R&D tax credits may primarily lead to wage increases for scientists rather than fostering innovation (Goolsbee, 1998; Romer, 2000). This underscores the importance of policy levers that can effectively boost the number of engineers and scientists in the long run.

This paper contributes to the literature exploring how the expansion of education can enhance innovation inputs (Andrews, 2023; Bianchi and Giorcelli, 2020; Carneiro et al., 2023; Kantor and Whalley, 2014). In contrast to most studies, this paper places particular emphasis on an accelerated education program that was only available during wartime and subsequently ceased to exist. Considering the challenges associated with scaling up college education, it is crucial to investigate the effects of accelerated education programs. The existing evidence in this domain remains scarce.

Lastly, this paper contributes to our understanding of the enduring consequences of the wartime transformation of the economy (Garin and Rothbaum, 2022; Giorcelli, 2023; Gross and Sampat, 2023; Jaworski, 2017; Kantor and Whalley, 2023; Schweiger et al., 2022). Compared to existing literature that primarily examines upper-tier knowledge production, this paper shifts its focus to a program designed for training relatively lower-skilled workers over a period of 12-16 weeks.

## 2. Institutional background

## 2.1 The sudden increase of STEM workers

The question of whether there is a shortage of engineers and scientists has been a subject of significant debate (Arrow and Capron, 1959; Blank and Stigler, 1957). Figure 1 indicates a notable increase in both the number of engineers and STEM workers per thousand population over the 20th century, highlighting a significant change in trend around the 1940s.

Between 1910 and 1940, the number of engineers per thousand population grew by an average of 0.51 per decade. In the same period, STEM workers increased by an average of 4.29 per decade. Post-1940, these rates notably increased. From 1940 to 2000, the average decadal increase for engineers was 1.76 per thousand population, while for STEM workers, it was substantially higher at 16.25 per decade. This demonstrates a more than threefold increase in the growth rate for engineers and nearly a fourfold increase for STEM workers after 1940 compared to before.

The sharp rise in the number of engineers in the 1950s was indeed surprising to many contemporaries. Blank and Stigler (1957) noted the unexpected surge in engineers around 1950, which was so significant that some questioned the accuracy of the 1950 census, suggesting it may have been erroneous in some manner. The data indeed reflect this sentiment, showing that the number of engineers per thousand jumped from 2.9 in 1940 to 5.68 in 1950, an increase that outpaced the growth seen in previous decades. The change in the growth rate of the share of STEM workers persisted up to 2000.

After evaluating multiple factors, Blank and Stigler (1957) concludes that the rise in the number of professional engineers is a reasonable estimate. They claim that several factors, including skill upgrading during WWII, the return of war veterans, and the presence of ESMWT training, could have contributed to this trend.

## 2.2 Engineering, Science, and Management War Training program

Between October 9, 1940, and June 30, 1945, approximately 1.8 million students received training through the Engineering, Science, and Management War Training (ESMWT) program. The program initially started as "Engineering Defense Training" and later changed its name to "Engineering, Science, and Management Defense Training" before finally becoming "Engineering, Science, and Management War Training." The federal government allocated \$60 million (equivalent to \$1315

million in 2024) to cover educational expenses for these students (Armsby, 1946, p.115). These courses were provided free of charge and were administered by colleges and universities under the sponsorship of the US Office of Education. The demand for additional engineers and scientists became increasingly apparent in the spring of 1940.

The educational content of the program encompassed a wide range of subjects, extending beyond science and engineering to include accounting, industrial organization, office management, personnel administration, procurement, statistics, and transportation. However, fields unrelated to war industries, such as finance, marketing, economic theory, sociology, and political science, were not incorporated into this program. In terms of course enrollment, electrical engineering had the highest number of participants, followed by general engineering and industrial engineering.

The individuals who received this training represented a diverse cross-section of the United States. This group included a wallpaper saleswoman, a receiving agent at a sugar firm, draftsmen from architectural firms, laboratory technicians, freelance artists, librarians, photographers, sculptors, gardeners, and counselors in horseback riding and riflery. The program typically lasted for 12-16 weeks, and training occurred not only on university campuses but also in non-eligible colleges, high schools, and industrial plants. For example, UCLA's training programs were primarily conducted in industrial plants located in Burbank, Long Beach, Wilmington, Inglewood, and Huntington Park, rather than on its university campus (Hamilton and Jackson, 1969, p.102). Data from North Carolina State College in 1942 indicates that 28.2 percent of the trainees held a high school diploma, 40.3 percent had completed 1 or 2 years of college education, and 25.2 percent had 3 or 4 years of college education (Ruggles, 1942).

Studies conducted by the Army Service Forces indicate that many WWII veterans expressed a desire to further their education but were reluctant to commit to a full four-year college program. Instead, they preferred courses that resembled the ESMWT curriculum, allowing them to quickly prepare for employment (Armsby, 1946, p.62).

This program resulted in a heightened interest in adult education and a widespread appreciation of effective teaching. Many individuals discovered their ability to learn from textbooks and lectures, broadening their horizons. Moreover, it contributed

to a greater recognition of night school and extension services offered by colleges. Additionally, it emphasized the significance of collaboration between universities and industries. Small firms often lacked the resources to provide training for their employees, but this program effectively trained a substantial workforce, leading to increased productivity (Armsby, 1946, p.65).

Universities gathered information on training requirements from their regional ESMWT representatives, while the War Manpower Commission conducted surveys to assess the needs of various industries. Identifying local training needs presented a significant challenge, as industries could not predict their war contracts in advance, leaving them uncertain about the specific terms of their wartime agreements.

University representatives maintained open lines of communication with both industries and the federal government, facilitating the exchange of crucial information. In many instances, the needs of businesses diverged from the conventional academic curriculum. Universities were tasked with departing from traditional academic norms, tailoring their courses to promptly address the economy's immediate demands, with a focus on practical education rather than general or technical subjects.

### **3. Data**

**ESMWT**—To quantify the intensity of ESMWT across commuting zones, we rely on Armsby (1946)'s data, which records the number of ESMWT trainees for each institution (p.115). We collect data on 1,795,716 trainees from 227 universities and colleges across the United States.

One drawback of this dataset is that it lacks information regarding the actual training locations. For example, the available data only provides information on the number of trainees who received training at the University of California. Yet, University of California conducted its training programs at multiple sites, including Berkeley, Los Angeles, and San Diego. Moreover, the majority of instructional sessions led by UCLA faculty took place at industrial plants in the Los Angeles area rather than on the university campus (Hamilton and Jackson, 1969, p.102).

To address this challenge, we turn to the catalog titled "Engineering Defense Training—Courses Approved to May 12, 1941," which we obtained from the National Archives at College Park. This catalog provides details about the precise train-



ing locations of each institution down to the town level.

For one exception, Pennsylvania State College, where the catalog only mentions “approximately 50 centers in central and western Pennsylvania,” we conducted a search for the joint frequency of “Pennsylvania State College” and “Engineering, Science, and Management War Training” on newspapers.com. This allowed us to ascertain whether a specific county was involved in the training or not. To distribute the trainees effectively, we employ population weights for each county. For instance, if University A trained a total of 60 workers across counties  $x$  and  $y$ , and county  $x$  has a population of 300 while county  $y$  has a population of 150, we assume that 40 trainees from University A were trained in county  $x$ , and the remaining 20 were trained in county  $y$ .

**Occupational and industrial structure**—We use 1% sample or 5% sample Census data covering the years from 1910 to 2000 sourced from Integrated Public Use Microdata Series (IPUMS). Our primary focus in this study revolves around two occupational groups: professional engineers and STEM workers. We identify these groups using the occupation classification defined by 1950 standards (occ1950) within the census data. For STEM workers, we adhere to Peri et al. (2015)’s definition of STEM workers and employ the broadest definition. We manually identify relevant occupations based on the 1950 definition from their definition of STEM occupations.

To identify the industries most impacted by the significant demand surge during WWII, we focused on manufacturing sectors that experienced the most substantial employment growth between 1940 and 1950. We categorized the top ten manufacturing industries with the highest employment growth as those related to war and defense.

**WWII controls**—To account for the impact of WWII labor demand shocks at the commuting zone level, this paper includes controls for various factors normalized by the 1940 population. Specifically, it incorporates the share of WWII veterans, war contracts, and war factories. The number of WWII veterans is derived from individual-level census data, while data on war supply contracts and war facilities is sourced from the County and City Data Book (ICPSR 7736).

**Public opinion poll**—To gauge attitudes toward education, we use public opinion poll data from Gallup in 1948, which we obtained from the Roper Center for Public Opinion Research. Specifically, we focused on the questionnaire that asked respondents, “if the colleges and universities of the country opened their doors for two full

weeks every year to anyone who wanted to come and study in any field, would you, yourself, be interested in taking the time and trouble to attend such a course?" This questionnaire encompassed nine distinct academic subjects, and for each subject, we employed an indicator as an outcome variable.

**Graduate dissertations**—To quantify the number of STEM students educated at each institution, we relied on the ProQuest Dissertations&Theses database. This database provides information on the name, title, institution, graduation year, and major of each graduate dissertation. We identified STEM majors if a paper is categorized under "Applied sciences", "Pure sciences", "Biological sciences", "Earth sciences", or "Health and environmental sciences"

**Patents**—To assess local invention rates, we use patent data from Petralia et al. (2016), which extends up to 1975. For patents granted after 1975, we incorporated data from PatentsView. In cases involving patents with multiple inventors, we assigned the location of the first inventor as the patent's location.

**Commuting zone crosswalk**—In our analysis, we allocated each data to commuting zones (CZ) by using the crosswalk file by Autor and Dorn (2013). For data in 1960, we utilized the crosswalk file from Koenig (2023). We restrict our attention to 722 commuting zones in the continental United States, excluding Alaska and Hawaii. More detail on the crosswalk process is explained in Online Appendix.

## 4. Empirical analysis

### 4.1 Geographical distribution of ESMWT training

First, we define the major treatment variable at the commuting zone level. Figure 2 presents a geographical distribution of the ESMWT trainees per thousand individuals across the United States' commuting zones. The varying shades of blue indicate the intensity of ESMWT trainees, with darker tones representing higher concentrations. The map reveals a heterogeneous distribution of ESMWT trainees, with some regions showing a much higher prevalence of trainees than others.

The commuting zone encompassing Bryan city, TX, holds the highest concentration, with an ESMWT share of 0.7064, marking it as a significant hub of wartime training activity. Following this, Wichita city, KS, with a share of 0.3180, and Decatur city, IL, with a share of 0.2164, are also prominent areas of ESMWT participation.

These specific zones might reflect an intensified engagement with the ESMWT program, possibly due to localized industrial endeavors pertinent to the World War II effort or particular socioeconomic frameworks that favored elevated levels of training.

## 4.2 Balance test

Table 1 examines the correlation between ESMWT program and local economic indicators. The table addresses the econometric concern of whether the intensity of ESMWT training is related to future local outcomes for reasons other than the training itself.

The table presents a series of regressions involving various variables and their relationship with the ESMWT training share. Each column adds incremental controls to the analysis. In Column (1), we establish the baseline by providing mean values for these variables. Moving to Column (2), we regress a set of variables linked to local economic potential against the ESMWT share, while also incorporating census region fixed effects. The regressions are weighted based on the 1940 population data.

This analysis reveals a discernible correlation between the ESMWT training share and several economic variables tied to World War II, such as supply contracts, wartime facilities, and the prevalence of veterans. This is consistent with statistical evidence from 1944 that highlighted the correlation between local wartime contracts and the proportion of trainees per capita (Bixler, 1944). Additionally, it is apparent that the ESMWT share is associated with other socioeconomic indicators, including the proportion of college graduates, engineers, and manufacturing value added.

To further investigate the economic impact, we examine income differentials using 1 percent individual census data from 1940. After controlling for race, gender, and birth year fixed effects, we find that a one standard deviation increase in ESMWT training intensity corresponds to a four percent increase in income in 1940.

The central premise underlying this study is that the variation in ESMWT training intensity across different regions is predominantly driven by the erroneous labor shortage predictions made by the War Manpower Commission's regional directors. These forecasts of labor shortages underwent regular revisions throughout World War II, and the allocation of ESMWT training was adjusted accordingly in response to these changing projections. This assumption is based on the inherent challenge of

accurately predicting labor market demands during wartime.

In essence, the assumption is that the allocation of training was influenced more by the imperfect forecasts of labor shortages rather than actual wartime labor supply conditions, which are measured using indicators such as the soldier draft rate or wartime supply contracts. The residual variation in training intensity, after accounting for these actual labor shortage indicators, can be attributed to the inaccuracies in the Commission's predictions.

To validate this assumption, column (3) extends the control variables to include WWII characteristics. It indicates that other economic characteristics tend to be evenly distributed across commuting zones once actual labor demand during WWII is taken into account, thereby attenuating concerns over potential endogeneity of ESMWT with future economic outcomes.

### 4.3 Commuting zone level evidence

To quantify the impact of the ESMWT on the occupational structure, we estimate the following equation.

$$y_{zt} = \gamma_t(ESMWT_z \times D_t) + \sum_k \phi_{kt}(X_{kz} \times D_t) + D_z + D_{r(z)t} + \varepsilon_{zt} \quad (1)$$

where  $y_{zt}$  represents the count of engineers or STEM workers per one thousand individuals in commuting zone  $z$  during census year  $t$ .  $ESMWT_z$ , which indicates the count of ESMWT trainees per one thousand individuals aged between 15-64 in the year 1940, is interacted with census year dummies  $D_t$ . To aid interpretation, the ESMWT share variable has been standardized, resulting in a mean of 0 and a standard deviation of 1. A one standard deviation increase in the ESMWT program share corresponds to an addition of 34.03 trainees per one thousand individuals within the population aged 15 to 64. Furthermore,  $X_{kz}$  includes a set of WWII-related characteristics, including war supply contracts, war facilities, and the presence of WWII veterans.  $D_z$  and  $D_{r(z)t}$  represent commuting zone and census region-year fixed effects, respectively. These fixed effects account for factors that do not vary over time and region-specific time trends. All these factors are adjusted based on the 1940 population size. Regressions are weighted by 1940 commuting zone population and standard errors are clustered by commuting zones.

The analysis of the impact of the ESMWT share on the density of engineers

per thousand reveals positive coefficients post-1940, reaching statistical significance from the 1960s forward. This pattern indicates no pre-existing trend prior to 1940, suggesting the observed effects are likely attributable to the ESMWT program itself. Over time, the magnitude of the ESMWT share's impact grows, hitting its stride in the 1970s and 1980s. This trend could reflect the program's delayed influence or the accumulated advantages of ongoing training. Specifically, our results show that an increase of one standard deviation in the ESMWT share (34.03 trainees per thousand population) led to an increase of 0.77 engineers per thousand by 1970.

In examining the STEM workforce, the findings align closely with those for engineers. The ESMWT share's coefficients are consistently positive and achieve statistical significance from the 1960s on, with the effect's strength intensifying in subsequent decades, peaking in the 1980s and 1990s. In quantitative terms, a one standard deviation rise in ESMWT share is linked to a rise of 3.7 STEM workers per thousand, underscoring the program's substantial impact on the STEM labor market. While the ESMWT training primarily focused on engineering, it is notable that its most significant long-term impact on local labor markets appears to be in STEM occupations beyond the field of engineering.

How do the estimates presented in Figure 3 translate into aggregate effects? We revisit the descriptive statistics in Figure 1. Between 1910 and 1940, STEM workers per thousand increased at a rate of 4.29 per decade. If this trend had continued until 2000, the number of STEM workers per thousand would have reached 49.52. However, the actual number of STEM workers per thousand in 2000 was 121.26. Therefore, there was an excess of 71.74 STEM workers per thousand in 2000 compared to the trend.

As shown in Figure 3, a one standard deviation increase in ESMWT training per thousand (34.03 trainees per thousand) led to an additional 3.49 STEM workers per thousand in 2000. The total number of ESMWT trainees was 1.8 million, and the total population aged between 15 and 64 in 1940 was 130 million. This results in 14.3 ESMWT trainees per thousand in 1940. Therefore, the effect of ESMWT accounts for approximately 2 percent of the excess STEM workers in 2000.

#### 4.4 Cross-cohort evidence

To gain a more detailed understanding of the impact of the ESMWT program, we delve into cohort-level variation. Data from a 1942 survey conducted at North Car-

olina State University (Ruggles, 1942, p. 844) reveals that the majority of ESMWT trainees were young, with approximately 75 percent falling within the 18 to 25 age bracket. This corresponds to birth years ranging from approximately 1917 to 1924. The remaining 23 percent of trainees, aged between 26 and 50, would have birth years ranging from approximately 1892 to 1916.

This approach allows us to gain a more granular understanding of how the program influenced different age groups within the population. We estimate the following equation:

$$y_{zbhft} = \gamma_t(ESMWT_z \times D_b) + \sum_k \phi_{kt}(X_{kz} \times D_b) + D_{zfh} + D_{r(z)fhb} + D_{r(z)fht} + \varepsilon_{zbhft} \quad (2)$$

where  $z$  represents the commuting zone,  $t$  denotes the census year,  $b$  stands for the birth cohort,  $h$  is an indicator for being white,  $f$  is an indicator for being female, and  $r(z)$  represents the census region of a commuting zone. The outcome variable  $y_{zbhft}$  signifies the number of engineers or STEM workers per thousand individuals in each cohort-gender-commuting zone bin.  $ESMWT_z$  and  $X_{kt}$  represent the intensity of ESMWT training and WWII characteristics, as previously described.

To account for potential confounding factors, we introduce commuting zone-gender-race fixed effects, census region-gender-race-birth cohort fixed effects, and census region-gender-race-census year fixed effects into our analysis. These higher-order fixed effects effectively absorb lower-order terms. Furthermore, these regressions are conducted with weighting by the population in each commuting zone-census year-birth cohort-race-gender group, and the standard errors are clustered by commuting zone.

This approach allows us to comprehensively examine the impact of ESMWT training on different demographic groups within commuting zones while considering multiple variables and controlling for various factors.

Figure 4 comprises two panels, (a) and (b), each depicting the estimated effect of the Engineering, Science, and Management War Training (ESMWT) program share on the number of engineers and STEM workers per thousand, respectively. In panel (a), the effect on engineers per thousand is shown over different birth year bins. The estimates suggest no significant effect around the time before and during the ESMWT program's early years (indicated by the vertical dashed line). Post the

ESMWT period, the coefficients turn positive, suggesting an increase in the number of engineers per thousand with each subsequent birth year bin. The error bars, representing confidence intervals, indicate increasing variability in the estimates as we move away from the program's inception.

Panel (b) illustrates the effect on STEM workers per thousand and displays a similar trend, with the coefficients remaining close to zero in the earlier bins and turning positive in the later periods. This panel shows a more consistent and pronounced positive trend, suggesting that the ESMWT program share may be more strongly associated with the number of STEM workers than engineers alone. The ascending trajectory of the coefficients, along with widening confidence intervals, indicates that the program's influence grows over time.

## 5. Mechanism

### 5.1 Change in attitude towards STEM education

In this section, we delve into the underlying mechanisms behind the impacts of ESMWT on local labor markets. Anecdotal evidence suggests that this program influenced people's attitudes towards education (Armsby, 1946, p.65). To explore the possibility of attitudinal changes, we utilize Gallup Poll surveys from 1948. Specifically, one of the questions posed to respondents inquired, "If the colleges and universities of the country opened their doors for two full weeks every year to anyone who wanted to come and study in any field, would you, yourself, be interested in taking the time and trouble to attend such a course?" We use this item as a metric to gauge the inclination towards studying a particular academic subject across states and estimate the following equation:

$$subject_i = \beta ESMWT_s + X_i' \gamma + \varepsilon_i \quad (3)$$

In the equation,  $subject_i$  is an indicator representing the academic subject a respondent is willing to study.  $ESMWT_s$  denotes the number of trainees per population aged between 15-64 at the state level. For ease of interpretation,  $ESMWT_s$  is standardized with a mean of 0 and a standard deviation of 1.  $X_i$  constitutes a vector of state-level or individual-level controls. State-level controls include WWII characteristics such as war contract, war facilities, and WWII veterans, all normalized by

the 1940 population. Individual controls include fixed effects accounting for gender, race, age, city size, household wealth, and the region of each state. Following Berinsky et al. (2011), we weight each individual by education weight (WtPubEd), which accounts for sampling bias in education and age.

Our results, as presented in Table 2, indicate that individuals living in states with greater ESMWT training intensity exhibited a heightened inclination to express interest in studying subjects such as mathematics, biology, and vocational training in electricity or television. The baseline probability of a respondent answering that she likes to study mathematics or biology for two weeks was 10 percent. However, a one standard deviation increase in ESMWT intensity was associated with a 5 percent increase in the probability of respondents choosing to study math or biology. Conversely, they showed less enthusiasm for pursuing studies in agriculture or home economics. Notably, no significant differences were observed in the remaining academic disciplines. Despite the lack of finer-grained data in the public opinion survey, it is reassuring to observe that the effects of ESMWT were primarily specific to STEM fields.

## 5.2 Effects on STEM graduate training

To investigate the mechanisms driving the increase in the proportion of STEM workers, we turn our attention to the educational institutions involved in the training. The availability of accelerated education programs may divert individuals who would have pursued more advanced degrees, potentially resulting in a net negative effect on the education system. To explore whether the increased interest in studying STEM-related subjects translated into a higher pursuit of advanced degrees, we collected data from the ProQuest Dissertations&Theses database, encompassing all master's or doctoral dissertations in STEM majors, and integrated this data with the ESMWT database we constructed. We then proceed to estimate the following equation:

$$dissertation_{nt} = \beta_t(ESMWT_n \times D_t) + D_n + D_{st} + \varepsilon_{nt} \quad (4)$$

In this equation, the outcome variable represents the count of dissertations awarded to graduate students in STEM fields in year  $t$  at institution  $n$ . The treatment intensity is defined as the natural logarithm of ESMWT trainees divided by the number of en-



engineering students in 1940, which we obtained from the 1940 edition of the *Journal of Engineering Education*. The terms  $D_n$  and  $D_{st}$  represent institution and state-year fixed effects, respectively. Regressions are weighted by the number of engineering students in 1940 and the error terms are clustered at the institution level.

Out of the 227 universities that participated in ESWMT, 105 institutions had both dissertation data and a presence of engineering students in 1939. To account for state-year fixed effects, we further restrict our sample to focus on a subset of 88 institutions.

Figure 5 displays the regression results. It appears that the increase in the interest for accelerated programs did not crowd out the interest in advanced skill formation as it led to an increase in the number of postgraduate students. Specifically, a one percent increase in the share of the training program at an academic institution is associated with a subsequent increase of 31 STEM graduates by 1990.

### 5.3 Effects on agglomeration spillover

The increase in the share of engineers could also have been driven by the persistent agglomeration of defense-related industries. To investigate this possibility, we revisit the regression equations from earlier sections. The following figures, (a) and (b), illustrate the transient effects of the ESMWT program on patents per capita and defense-related industries, respectively. Figure (a) utilizes Equation (1) with log patents per capita as the outcome variable, while Figure (b) employs Equation (2) with the share of workers in defense-related industries as the outcome variable.

We delineates defense-related industries based on the expansion of industry employment between 1940 and 1950. Using census data and 1950 classification of industries, the number of workers within each industry was tallied to compute growth rates. This analysis identified the top ten manufacturing industries exhibiting the most substantial growth over the decade.<sup>1</sup> The proportion of the workforce employed in these sectors serves as an indicator of the impact on industries that experienced the most pronounced demand shocks during World War II.

Panel (a) of Figure 6 shows the effect on log patents per capita. The plotted coef-

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<sup>1</sup>The industries, as denoted by their ind1950 codes, include 358 (Miscellaneous machinery), 399 (Miscellaneous manufacturing industries), 367 (Electrical machinery, equipment, and supplies), 377 (Aircraft and parts), 458 (Miscellaneous paper and pulp products), 346 (Fabricated steel products), 347 (Fabricated nonferrous metal products), 387 (Photographic equipment and supplies), 419 (Miscellaneous food preparations and kindred products), and 467 (Drugs and medicines).

ficients, which are derived from commuting zone by year level panel data, oscillate around zero without displaying a strong or consistent trend. The error bars, representing confidence intervals, are relatively wide, suggesting considerable uncertainty in these estimates. The lack of a sustained positive or negative effect across the birth year bins indicates that the ESMWT program's impact on patents per capita was not enduring and does not support the hypothesis that the program contributed to a long-term agglomeration of inventors.

Panel (b) presents the effect on defense-related industries, estimated using a more granular panel data set that uses commuting zone, birth year, year, race, and gender cells. Similar to panel (a), the coefficients in panel (b) are not consistent over time, peaking in the mid-period bins before returning to near zero. This pattern suggests that any impact of the ESMWT program on employment within defense-related industries was also not persistent over time.

The information depicted in both panels helps rule out alternative explanations such as the long-term agglomeration of inventors or the sustained concentration of defense-related industries as a result of the ESMWT program. The transient nature of the effects, as indicated by the return to near-zero coefficients in later years, suggests that the influence of the ESMWT program is unlikely to be explained by agglomeration.

## 6. Conclusion

This paper examines the long-term impact of ESMWT on the growth of engineers and STEM workers. Controlling for WWII characteristics, the variation in ESMWT training across commuting zones is primarily driven by idiosyncratic predictions of imminent labor scarcity during WWII.

The results indicate that this training program led to a lasting increase in the number of engineers and STEM workers. In aggregate, ESMWT accounts for 2 percent of the excess STEM workers in 2000. While this may not appear particularly substantial, it is noteworthy considering the program's short duration of just five years and its limited reach, with only about 1 percent of the general population receiving the training. Furthermore, evidence from graduate dissertations shows that the program did not deter individuals from pursuing advanced training in the future, highlighting its positive long-term effects.

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Table 1: Balance test

|                                     | (1)    | (2)                       | (3)                                         | (4)     |
|-------------------------------------|--------|---------------------------|---------------------------------------------|---------|
|                                     | Mean   | ESMWT share<br>+region FE | ESMWT share<br>+region FE<br>+WWII controls | Obs     |
| <i>WWII controls:</i>               |        |                           |                                             |         |
| WWII supply contracts (\$k per pop) | 0.42   | 0.26(0.14)*               |                                             | 722     |
| WWII veterans share 1950            | 0.1    | 0.01(0.002)**             |                                             | 722     |
| WWII facilities (\$k per pop)       | 0.19   | 0.01(0.01)                |                                             | 722     |
| <i>Socioeconomic controls:</i>      |        |                           |                                             |         |
| Engineers per thousand 1940         | 1.73   | 0.22 (0.13) *             | -0.03(0.06)                                 | 722     |
| STEM workers per thousand 1940      | 18.21  | 1.15(0.65) *              | -0.12 (0.25)                                | 722     |
| College graduates per thousand      | 98.12  | 3.92(1.58)**              | 1.28(1.10)                                  | 722     |
| Mfg. value added 1940 (\$k per pop) | 0.07   | 0.02 (0.01) *             | 0.01 (0.01)                                 | 722     |
| Urban share 1940                    | 0.32   | 0.04 (0.03)               | 0.02 (0.01)                                 | 722     |
| (log) population                    | 122845 | 0.28(0.16)*               | 0.15(0.10)                                  | 722     |
| (log) patents per thousands         | 1.94   | 0.06(0.08)                | 0.004(0.04)                                 | 722     |
| (log) weekly wage (\$) 1940         | 15.54  | 0.04(0.02)*               | 0.01(0.01)                                  | 966,804 |

*Notes:* This table presents the outcomes of regressing ESMWT share variables (standardized with a mean of 0 and a standard deviation of 1) against a set of variables. In column (2), we introduce census region fixed effects, while in column (3), we incorporate WWII characteristics alongside census region fixed effects. The unit of analysis comprises 722 commuting zones within the continental United States. The final row examines cross-commuting zone variations in individuals' wages. In addition to commuting zone controls, it also controls for race, gender, and birth year fixed effects. The results at the commuting zone level are weighted by the 1940 population, while the individual-level results are weighted by the census sampling weight. Robust standard errors in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$

Table 2: Willingness to study

| Dep var:     | (1)              | (2)             | (3)               | (4)                | (5)            | (6)             | (7)              | (8)             | (9)               |
|--------------|------------------|-----------------|-------------------|--------------------|----------------|-----------------|------------------|-----------------|-------------------|
|              | math             | vocation        | agriculture,      | home               | bookkeeping,   | business        | law,             | social          | fine art          |
|              | biology          | electricity     | gardening         | economics          | stenography    |                 | medicine         | science         |                   |
| esmw_t_share | 0.05**<br>(0.02) | 0.03*<br>(0.01) | -0.04**<br>(0.02) | -0.05***<br>(0.02) | 0.01<br>(0.01) | -0.02<br>(0.02) | -0.003<br>(0.01) | -0.01<br>(0.02) | -0.005<br>(0.005) |
| Constant     | 0.10***          | 0.11***         | 0.08***           | 0.14***            | 0.03***        | 0.12***         | 0.06***          | 0.16***         | 0.03***           |
|              | (0.01)           | (0.01)          | (0.01)            | (0.01)             | (0.004)        | (0.01)          | (0.01)           | (0.01)          | (0.004)           |
| Controls     | ✓                | ✓               | ✓                 | ✓                  | ✓              | ✓               | ✓                | ✓               | ✓                 |
| State        | 42               | 42              | 42                | 42                 | 42             | 42              | 42               | 42              | 42                |
| Observations | 979              | 979             | 979               | 979                | 979            | 979             | 979              | 979             | 979               |
| R-squared    | 0.10             | 0.15            | 0.23              | 0.18               | 0.12           | 0.12            | 0.10             | 0.09            | 0.08              |

Notes: The outcome variable in this analysis is an indicator for respondents' interest in pursuing academic subjects. It is based on their responses to the question, "If the colleges and universities of the country opened their doors for two full weeks every year to anyone who wanted to come and study in any field, would you, yourself, be interested in taking the time and trouble to attend such a course?" from the 1948 Gallup Poll. The treatment variable, denoted as ESMWT, is defined as the number of ESMWT trainees divided by the population in 1940 at the state level. Our control variables include WWII supply contracts, WWII facilities, WWII veterans (normalized by the 1940 population) at the state level, as well as fixed effects for gender, race, age, city size, household wealth, and the region within each state at the individual level. Each individual is weighted by their education weight (WtPubEd). Robust standard errors in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$

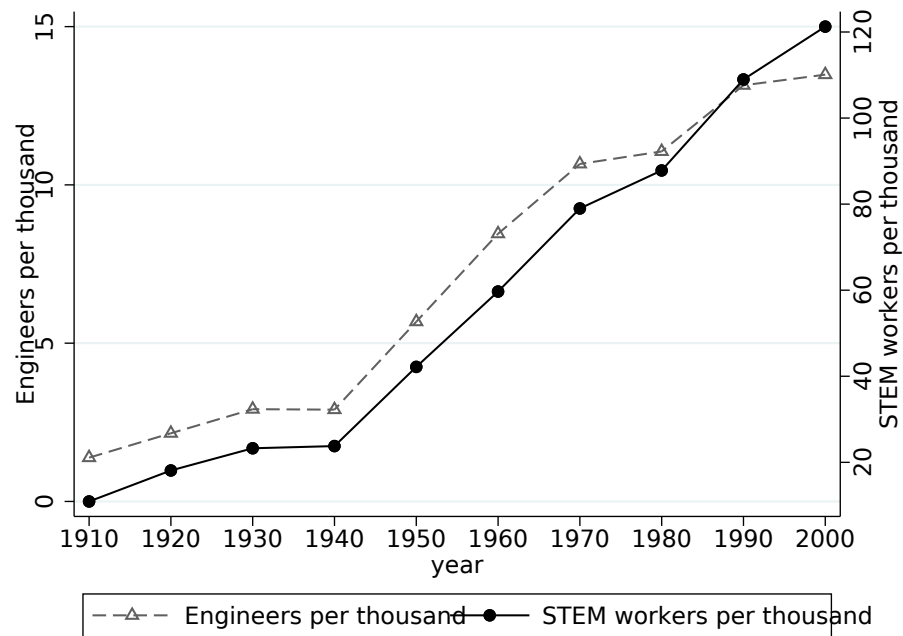


Figure 1: Total number of engineers and STEM workers per thousand

*Notes:* The figure represents the number of professional engineers and STEM workers per thousand people aged between 15 and 64. We map Peri et al. (2015)'s broadest definition of STEM workers into occupation code based on 1950 definitions.



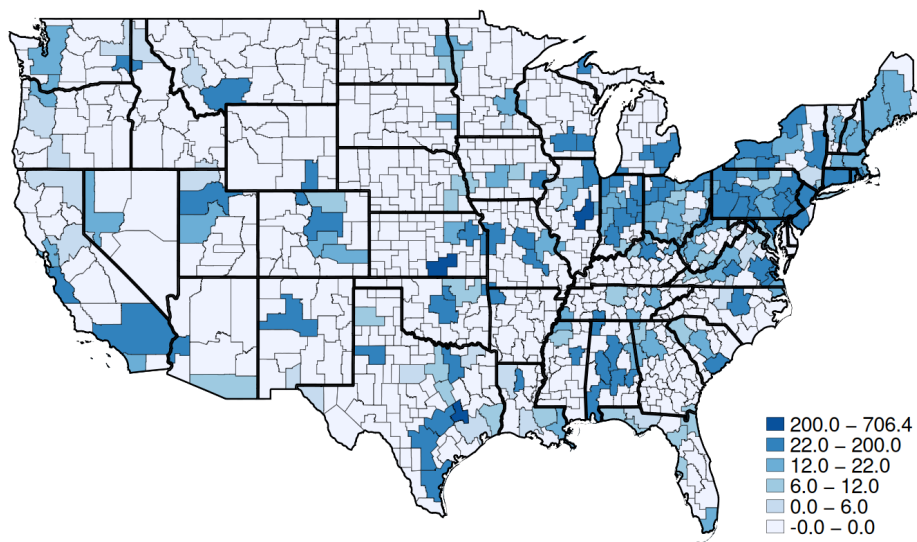
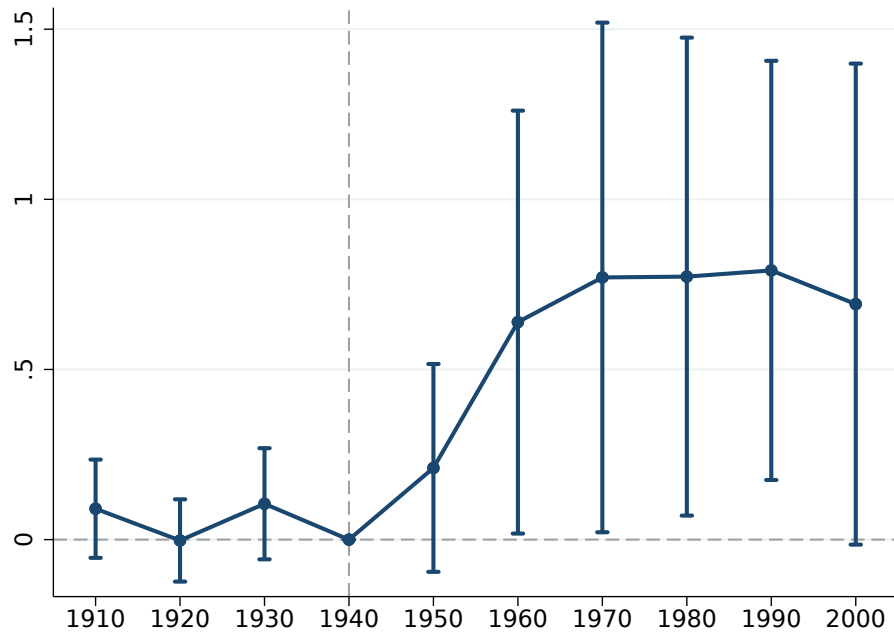
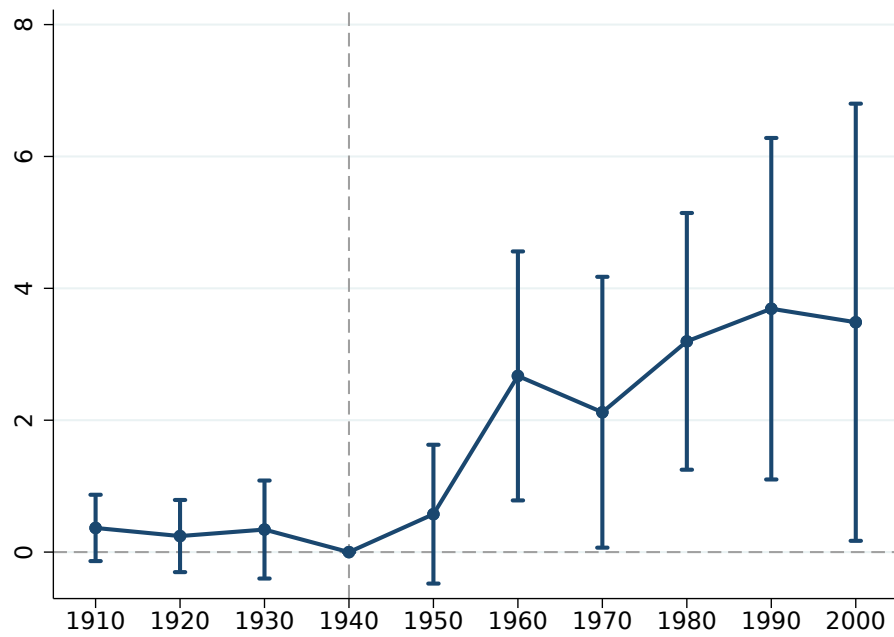


Figure 2: ESMWT trainees per thousand

*Notes:* This figure depicts the geographical distribution of ESMWT trainees across commuting zones. The number of trainees from each university is allocated to multiple commuting zones using actual training location data obtained from a catalog at the National Archives. In cases where trainees are assigned to multiple commuting zones, we distribute them based on population weight.



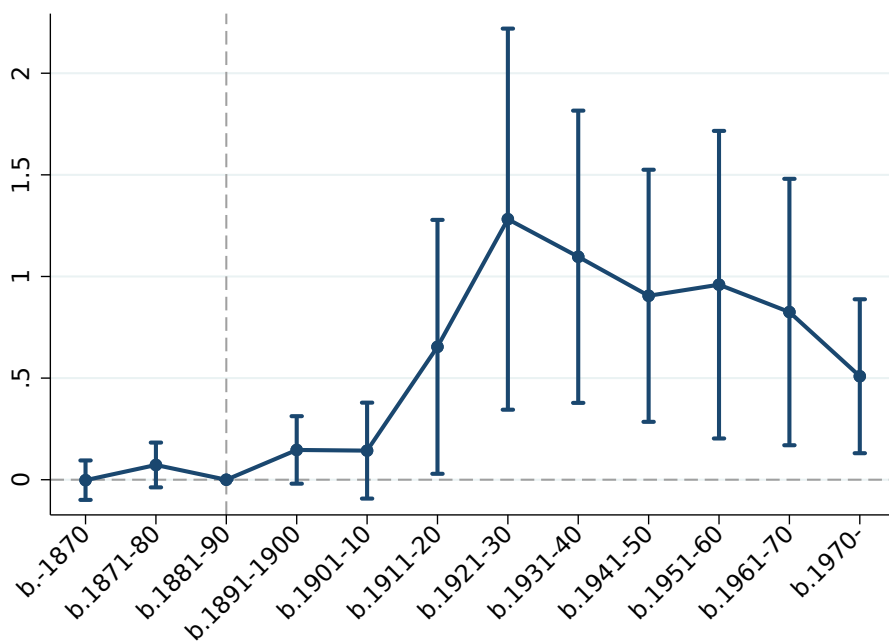
(a) Engineers per thousands



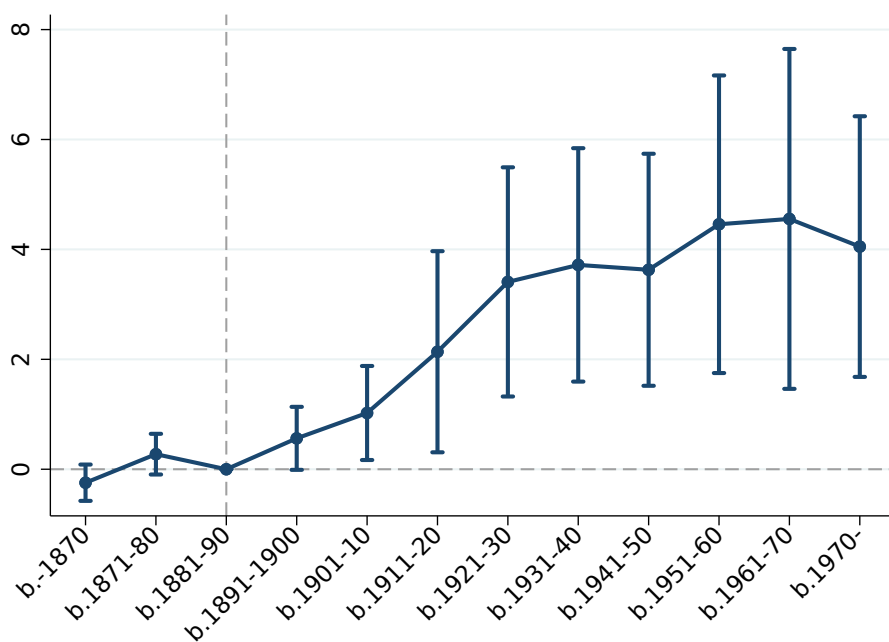
(b) STEM workers per thousands

Figure 3: Effect of ESMWT training on occupational structure at the commuting zone level

*Notes:* This figure presents the regression results from Equation (1). The outcome variables, representing the number of professional engineers or STEM workers per thousand workers, are regressed on ESMWT share interacted with time dummies. The control variables include WWII supply contracts, WWII facilities, and WWII veterans, all interacted with time dummies, as well as commuting zone fixed effects and census region by year fixed effects. 95 percent confidence intervals are plotted with vertical lines.



(a) Engineers per thousands



(b) STEM workers per thousands

Figure 4: Effect of ESMWT training on the occupational structure

*Notes:* This figure presents the regression results from Equation (2). The outcome variables, representing the number of professional engineers or STEM workers per thousand workers, are regressed on ESMWT share interacted with cohort dummies. The control variables include WWII supply contracts, WWII facilities, and WWII veterans, all interacted with cohort dummies, as well as commuting zone-gender-birth cohort fixed effects, census region-gender-race-birth cohort fixed effects, and census region-gender-race-census year fixed effects. 95 percent confidence intervals are plotted with vertical lines.

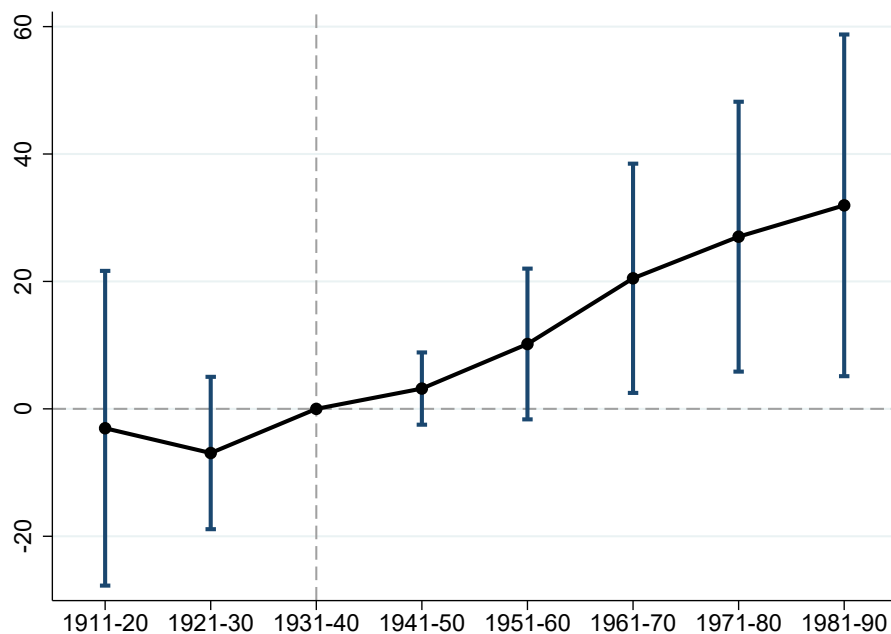
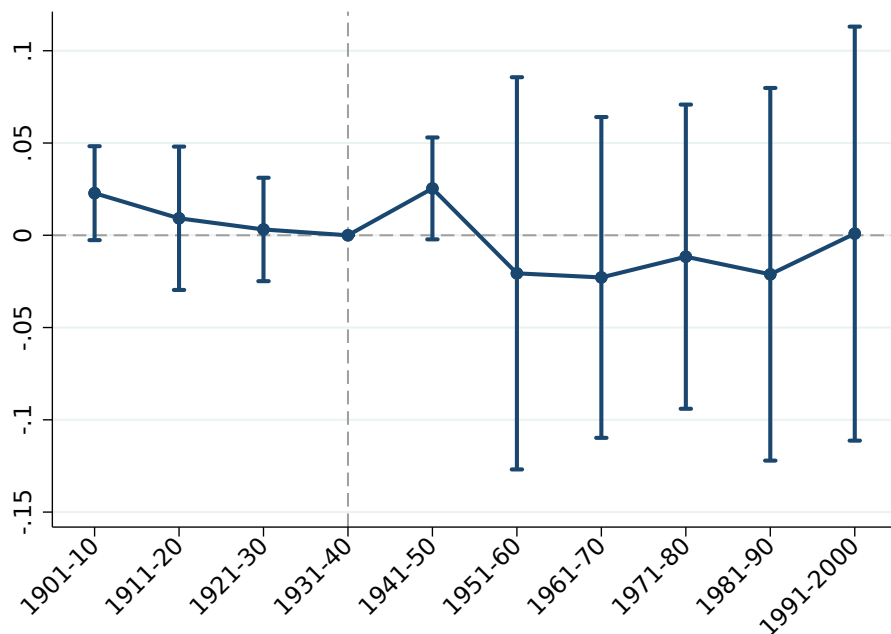
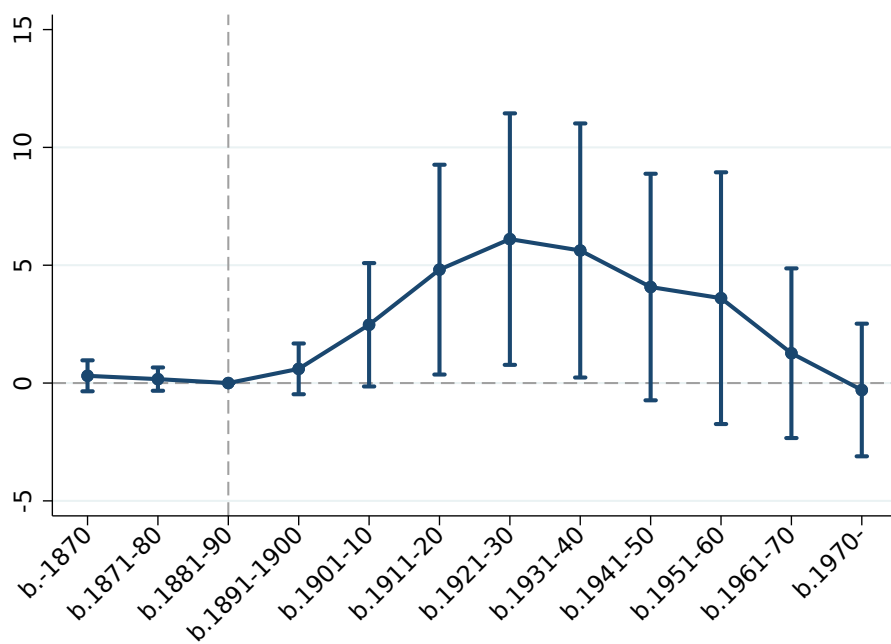


Figure 5: Effect on post-graduate education

*Notes:* This figure shows the regression outcomes derived from Equation Equation (4). The number of post graduate students at each academic institutions are regressed on log of ESMWT trainees over engineering students in 1939 interacted with time dummies. The model accounts for institution and state-year fixed effects and is weighted by the number of engineering students in 1939. The 95 percent confidence intervals are depicted using vertical lines.



(a) Effect on log patents per capita



(b) Effect on defense-related industries

Figure 6: Effects on patents and defense-related industries

Notes: Figure (a) runs Equation (1), with the outcome variable being the logarithm of patents per capita in each commuting zone. Figure (b) runs Equation (2), with the outcome variables representing the share of workers in defense-related industries. The 95 percent confidence intervals are depicted using vertical lines.

## A. Appendix

### A.1 Data appendix

EDT-Misc. 35

Federal Security Agency  
U. S. OFFICE OF EDUCATION  
Washington

ENGINEERING DEFENSE TRAINING  
Courses Approved to May 12, 1941

CAPITALS indicate FULL-TIME courses; lower case, part-time. The city first listed is the location of the institution. If instruction is given at other points these follow in parentheses. Subjects marked with an asterisk (\*) are available in more than one city. Starting dates vary widely and many courses are repeated from time to time; for more detailed information apply directly to the institution. Courses that have been approved but subsequently cancelled are omitted from this list.

#### ALABAMA

|                                              |                                                            |
|----------------------------------------------|------------------------------------------------------------|
| <u>Alabama Polytechnic Institute, Auburn</u> | (Birmingham, Jacksonville, Mobile, Montgomery, Wilson Dam) |
| Construction Engineering*                    | Materials Handling                                         |
| Drafting Room Practice                       | Military Highway Engineering                               |
| Engineering Drawing*                         | Naval Architecture                                         |
| Engineering Economy                          | Production Engineering*                                    |
| Forge Shop Management                        | Route Surveying                                            |
| Graphics of Machines                         | Production Supervision                                     |
| Industrial Relations                         | Safety Engineering                                         |
| Kinematics of Machines                       | Slide Rule                                                 |
| Machine Design*                              | Surveying*                                                 |
| Machine Shop Management                      | Tool Engineering*                                          |
| Marine Engineering                           |                                                            |
| Marine Machinery                             |                                                            |

Figure A1: Engineering Defense Training courses approved

*Notes:* This data is used to allocate the number of trainees for each university across multiple commuting zones. The physical copy of this document was acquired from the National Archives at College Park (Record Group 12).

Table A1: IPUMS USA individual sample

| Year | Sample                 |
|------|------------------------|
| 1910 | 1% sample              |
| 1920 | 1% sample              |
| 1930 | 1% sample              |
| 1940 | 1% sample              |
| 1950 | 1% sample              |
| 1960 | 5% sample              |
| 1970 | 1% Form 1 Metro sample |
| 1980 | 5% State sample        |
| 1990 | 5% State sample        |
| 2000 | 5% sample              |

Table A2: Crosswalk process

| Decade                                          | Identifier                | Crosswalk                                             |
|-------------------------------------------------|---------------------------|-------------------------------------------------------|
| Decennial census 1910-1950                      | State Economic Area       | Autor and Dorn (2013)                                 |
| Decennial census 1960                           | Public Use Microdata Area | Koenig (2023)                                         |
| Decennial census 1970-1980                      | County Group              | Autor and Dorn (2013)                                 |
| Decennial census 1990-2000                      | Public Use Microdata Area | Autor and Dorn (2013)                                 |
| County and City Data Book<br>(ICPSR 7736)       | County                    | IPUMS, Autor and Dorn (2013)<br>(county to SEA to CZ) |
| Patent<br>(Petrulia et al. (2016), PatentsView) | County                    | IPUMS, Autor and Dorn (2013)<br>(county to SEA to CZ) |

## A.2 Robustness test

This subsection assesses the robustness of the main results by incorporating additional WWII characteristics: Army enlistment, Army deaths, and Navy deaths. These factors may be correlated with regional labor shortages. The data for these variables were sourced from Rose (2018).

Reassuringly, as shown in Table A3, the additional characteristics are not correlated with ESMWT share. The inclusion of these extra controls does not change the main results, as confirmed in both Table A4 and Table A5.

Table A3: Additional balance test

| Variables                           | mean  | difference<br>+region FE |
|-------------------------------------|-------|--------------------------|
| <i>WWII controls:</i>               |       |                          |
| WWII supply contracts (\$k per pop) | 0.42  | 0.26 (0.14)*             |
| WWII veterans share 1950            | 0.10  | 0.01 (0.003)**           |
| WWII facilities (\$k per pop)       | 0.19  | 0.01 (0.01)              |
| <i>Additional WWII controls:</i>    |       |                          |
| Navy deaths per thousand            | 0.54  | 0.004 (0.01)             |
| Army deaths per thousand            | 2.52  | 0.03 (0.05)              |
| Army enlistment per thousand        | 61.19 | 1.50 (1.03)              |

*Notes:* WWII controls are taken from County and City Data Book (ICPSR 7736). Additional WWII controls are taken from Rose (2018). Variables are regressed on ESMWT share variables along with census region fixed effects. Regressions are weighted by 1940 population. Robust standard errors are in parentheses. \*\*\* p<0.01, \*\* p<0.05, \* p<0.1



Table A4: Commuting zone level evidence—robustness check

|                                  | (1)                    | (2)                       | (3)                       | (4)               |
|----------------------------------|------------------------|---------------------------|---------------------------|-------------------|
| Dep var:                         | Engineers per thousand | STEM workers per thousand | STEM workers per thousand |                   |
| ESMWT share $\times I(t > 1940)$ | 0.60**<br>(0.26)       | 0.60**<br>(0.27)          | 2.39***<br>(0.88)         | 2.41***<br>(0.91) |
| CZ FE                            | ✓                      | ✓                         | ✓                         | ✓                 |
| Region-year FE                   | ✓                      | ✓                         | ✓                         | ✓                 |
| WWII controls                    | ✓                      | ✓                         | ✓                         | ✓                 |
| Additional WWII controls         |                        | ✓                         |                           | ✓                 |
| Commuting zones                  | 722                    | 722                       | 722                       | 722               |
| Observations                     | 7,220                  | 7,220                     | 7,220                     | 7,220             |
| R-squared                        | 0.88                   | 0.88                      | 0.97                      | 0.97              |

Notes: The results are based on Equation (1). The WWII controls consist of WWII supply contracts, WWII veterans, and WWII facilities, all normalized by the 1940 population, and interacted with time dummies. Additional WWII controls include Army deaths, Navy deaths, and Army enlistment during WWII, which have been normalized by the population and interacted with time dummies. These variables are sourced from Rose (2018).  $I(t > 1940)$  is an indicator variable which takes the value of 1 for census years after 1940. Regressions are weighted by commuting zone population in 1940. Standard errors are clustered by commuting zone. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$

Table A5: Cross-cohort evidence—robustness check

|                                  | (1)                    | (2)                       | (3)               | (4)               |
|----------------------------------|------------------------|---------------------------|-------------------|-------------------|
| Dep var:                         | Engineers per thousand | STEM workers per thousand |                   |                   |
| ESMWT share $\times I(b > 1891)$ | 0.79***<br>(0.25)      | 0.73***<br>(0.25)         | 3.45***<br>(0.89) | 3.33***<br>(0.85) |
| CZ-gender-birth cohort FE        | ✓                      | ✓                         | ✓                 | ✓                 |
| region-gender-race-cohort FE     | ✓                      | ✓                         | ✓                 | ✓                 |
| region-gender-race-year FE       | ✓                      | ✓                         | ✓                 | ✓                 |
| WWII controls                    | ✓                      | ✓                         | ✓                 | ✓                 |
| Additional WWII controls         |                        | ✓                         |                   | ✓                 |
| Commuting zones                  | 722                    | 722                       | 722               | 722               |
| Observations                     | 153,763                | 153,763                   | 153,763           | 153,763           |
| R-squared                        | 0.78                   | 0.78                      | 0.84              | 0.84              |

*Notes:* The results are based on Equation (2). The WWII controls consist of WWII supply contracts, WWII veterans, and WWII facilities, all normalized by the 1940 population, and interacted with cohort dummies. Additional WWII controls include Army deaths, Navy deaths, and Army enlistment during WWII, which have been normalized by the population and interacted with cohort dummies. These variables are sourced from Rose (2018).  $I(b > 1890)$  is an indicator variable which takes the value of 1 for cohorts born after 1890. Regressions are weighted by the aggregated sampling weights in each commuting zone-gender-race-birth cohort-census year bin. Standard errors are clustered by commuting zone. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$